THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
013 GEOGRAPHY
MARKING SCHEME NOVEMBER 2025

SECTION A (15 MARKS)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
D	B	D	B	A	B	A	B	A	B

2.

i	ii	iii	iv	v
A	H	F	E	C

SECTION B (70 MARKS)

3. (a). land cools down faster than the sea, creating cooler, denser air over the land and warmer, less dense air over the sea. This difference in temperature and density results in high pressure over the land and low pressure over the water, causing the cooler land air to flow offshore towards the warmer, low-pressure area over the sea

(b) land heating faster than water, creating warmer, less dense air over the land that rises. Cooler, denser air from the sea then flows in to replace the rising air, producing a breeze from the sea towards the land. This circulation is driven by the difference in temperature and thus pressure between the rapidly heating land and the slower-to-heat water

(c) is a diurnal (daily cycle) wind that blows up a mountain valley during the day. It is caused by solar heating, where the mountain slopes warm faster than the valley floor, causing the warm air on the slopes to rise, which in turn pulls cooler, denser air from the valley to flow uphill to replace it. This daily wind system is opposite to the mountain wind which blows down the valley at night when the slopes cool faster than the valley floor

(d) is a measure of water vapor in the air. Meteorologists typically describe water vapor in the atmosphere in two different ways: absolute humidity and relative humidity. Dew point is an absolute measure and helps us understand how muggy and humid it feels outside

4.

(a).

1HOUR =15 Degree
=123 degree

$$\frac{123 \times 1 \text{hour}}{15}$$

8hours and 12minutes

The local time will be 8;48 am

(b). 1hour =15degrees
=129 degrees

$$\frac{1 \text{hour} \times 129}{15}$$

8hours 36minutes

The local time will be 08:24 am

5. (a). total annual rainfall will be
175+170+160+184+180+179+177+100+138+90+112+130
The total Rainfall is 1795mm

(b). range means = largest – lowest
28.4-23=5.4℃

(c). $\frac{24.5+25.7+27.7+28.4+30+24+24+26+26+24+23+24}{12} = 257.3$

The annual mean is 21.44℃

(d). **High and Consistent Temperatures:**
Temperatures remain warm throughout the year, typically between 25-30°C (77-86°F), with very little difference between the hottest and coolest months.
Abundant and Even Rainfall:
High levels of precipitation, generally over 2000 mm (80 inches) annually, are spread evenly across the year, with no distinct wet or dry seasons.
High Humidity:
The combination of heat and moisture creates consistently high humidity levels, often exceeding 80%.

6. **Latitude**:
Areas closer to the equator receive more direct sunlight, resulting in higher temperatures, while polar regions receive less direct sunlight, leading to lower temperatures.
Altitude:
Higher altitudes have lower temperatures because the air is thinner and absorbs less heat from the sun.
Distance from the Sea:
Coastal areas have more moderate temperatures than inland areas because water heats up and cools down more slowly than land, moderating coastal temperatures.

7. (a)The Grid Method
   i. **How it works:** Overlay a grid of squares onto the map's irregular shape.
   ii. **Process:** Count the number of whole squares the shape covers, and estimate the area of the partial squares (e.g., count two half-squares as one).
   iii. **Calculation:** Multiply the total number of squares (full and partial) by the area of a single grid square (determined by the map's scale) to get the total area.

(b)
   i. The Division Method
   - **How it works:** Divide the irregular shape into smaller, simpler, regular geometric shapes.
   - **Process:** Break the irregular shape into shapes like triangles, rectangles, or squares.
   - **Calculation:**
   Calculate the area of each individual regular shape using its specific formula (e.g., area of a triangle = 0.5 * base * height) and then sum all the individual areas to find the total area of the irregular shape.

   ii. The Planimeter Method
   - **How it works:** A planimeter is a tool that can be used to directly measure the area of an irregular shape on a map.
   - **Process:** You manually trace the boundary of the irregular shape with the planimeter's measuring wheel.
   - **Calculation:** The device mechanically integrates the movements of its wheel to calculate

and display the exact enclosed area, eliminating the need for separate calculations

(c). i. Navigation and Route Planning:

- Maps show roads, landmarks, and other features, helping people plan journeys and find their way from one location to another.
- They are essential for travel by land, sea, and air to plot courses and maintain direction.

ii. General Reference and Information:

- Maps provide information about physical features like mountains and rivers, as well as political boundaries of countries, states, and cities.
- They are used to locate specific places, understand the layout of an area, and find points of interest.

iii. Regional Planning and Analysis:

- Urban and regional planners use maps to identify suitable areas for development, locate natural resources, and plan transportation networks.
- Different types of thematic maps show specific data, such as climate conditions, population distribution, or industrial areas, for analytical purposes

8.

i. **Ships disappearing over the horizon:** As a ship sails away, it appears to sink below the horizon, with the hull disappearing first, followed by the mast. On a flat Earth, the entire ship would simply become smaller but remain fully visible.

ii. **Lunar eclipses:** During a lunar eclipse, the Earth passes between the sun and the moon, casting its shadow on the moon. This shadow is consistently round and curved, a characteristic of a spherical object.

iii. **Photographs from space:** Satellite imagery and photos taken from space clearly show Earth as a sphere, providing direct visual evidence of its round shape.

iv. **Visible stars change with location:** The constellations visible in the night sky change depending on one's latitude. For example, different stars become visible as you travel north or south, and some stars are only visible in certain hemispheres.

v. **Circumnavigation:** Explorers like Ferdinand Magellan have sailed around the Earth, returning to their starting point by traveling in a consistent direction, such as west. This is only possible on a spherical body.

vi. **Time zones:** The Earth's rotation causes different parts of the planet to face the sun at different times, resulting in day and night and the existence of time zones. This variation in daylight and darkness across the globe is consistent with a spherical Earth.

9. (a)

Personal Safety and Disaster Preparedness: Forecasts provide early warnings of dangerous conditions like heatwaves, storms, and blizzards, allowing people to take necessary precautions, which can be a matter of life and death.

Economic and Business Operations: Weather forecasts help businesses plan for power production and consumption, manage transportation logistics, and avoid significant losses from unexpected weather events

Agriculture Optimization: Farmers and gardeners use forecasts to schedule irrigation, protect crops from frost, and make better decisions about planting and harvesting, which contributes to food security.

Informed Decision-Making for Activities: Individuals can plan outdoor activities and prepare for daily weather conditions, such as knowing to bring an umbrella on a rainy day or planning for a large outdoor event around a storm.

Resource Management and Public Works: Public entities and utility companies use forecasts to prepare for and respond to weather hazards. For example, **Tanzania Meteological Agency** mentions that power companies deploy personnel, and public works departments order salt and de-icers for winter storms.

(b)

Site Location and Surroundings:
The station should be located away from obstructions like buildings, trees, and steep slopes, which can affect wind, air temperature, and other measurements. For example, a thermometer must be protected from direct sunlight and given plenty of ventilation.
Power Supply:
Weather stations require a reliable power source. A separate, independent DC power supply, often from a solar panel, is recommended over using AC power, as it can reduce the risk of short circuits during heavy rains.
Sensor Placement and Height:
Instruments must be installed at the correct heights to avoid being blocked by obstructions. For example, a wind sensor might need to be on a 10-meter tower, and a temperature sensor should be no closer than a certain distance (e.g., four times its height) from large paved areas.
Data Logger Enclosure:
The data logger, the central unit for collecting and storing data, needs to be housed in a protective enclosure with satisfactory ingress protection. This safeguards the sensitive electronics from extreme weather conditions and potential water damage.
Instrument Accuracy and Calibration:
Sensors like thermometers, humidity sensors, and anemometers are the foundation of the station. It's crucial to ensure these instruments are accurate and calibrated to collect reliable data that reflects true atmospheric conditions.

SECTION C (15 MARKS)

10. **Causes the Four Seasons:**
The most significant effect of Earth's revolution, combined with its axial tilt, is the creation of the four seasons: spring, summer, autumn (or fall), and winter. Different parts of the Earth receive varying amounts of direct sunlight throughout the year, leading to different temperature patterns.
Variation in the Length of Days and Nights:
As Earth revolves around the sun, the length of days and nights varies in different parts of the Earth at different times of the year. For example, summer has longer days, while winter has longer nights.
Changes in the Position of the Midday Sun:
The apparent position of the midday sun in the sky changes throughout the year due to Earth's revolution. This causes the sun's rays to fall more directly on some parts of Earth at different times, influencing temperatures and weather patterns.
Occurrence of Solstices and Equinoxes:
The Earth's revolution also leads to solstices and equinoxes. Solstices mark the longest and shortest days of the year, occurring when one hemisphere is tilted most directly toward or away from the sun. Equinoxes occur when the sun shines directly on the equator, resulting in approximately equal day and night lengths across the global